

**Relational graph convolutional networks for predicting
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Relational graph convolutional networks for predicting blood-brain barrier penetration of drug molecules

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Abstract

Motivation: Evaluating the blood-brain barrier (BBB) permeability of drug molecules is a critical step in brain drug development. Traditional methods for the evaluation require complicated *in vitro* or *in vivo* testing. Alternatively, *in silico* predictions based on machine learning have proved to be a cost-efficient way to complement the *in vitro* and *in vivo* methods. However, the performance of the established models has been limited by their incapability of dealing with the interactions between drugs and proteins, which play an important role in the mechanism behind the BBB penetrating behaviors. To address this limitation, we employed the relational graph convolutional network (RGCN) to handle the drug-protein interactions as well as the properties of each individual drug.

Results: The RGCN model achieved an overall accuracy of 0.875, an AUORC of 0.810 and an AUPRC of 0.810 for the testing dataset with the drug-protein interactions and the Mordred descriptors as the input. Introducing drug-drug similarity to connect structurally similar drugs in the graph further improved the testing results, giving an overall accuracy of 0.881, an AUROC of 0.917 and an AUPRC of 0.818. In addition, the drug embeddings from the drug repurposing knowledge graph (DRKG), which encode the relations between drugs and various pharmacological entities, were found to be helpful in predicting BBB permeability. The AUROC and the AUPRC increased to 0.928 and 0.840, respectively, when the Mordred descriptors were combined with the DRKG embeddings. Our model is expected to provide high-confidence predictions of BBB permeability for drug prioritization in the experimental screening of BBB-penetrating drugs.

Availability and Implementation: The data and the codes are freely available at <https://github.com/dingyan20/BBB-Penetration-Prediction>.

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1. Introduction

Brain targeting drug development has long been suffering from poor penetration of drug molecules across the blood-brain barrier (BBB)¹⁻³. The BBB, mainly formed by a monolayer of tightly packed endothelial cells, is a physical, metabolic and transport barrier that strictly regulates the transfer of substances between the blood and the neural tissues⁴⁻⁶. Due to this barrier, most drug molecules cannot enter the brain, thus showing no therapeutic effects on central nervous system

(CNS) disorders. Therefore, evaluating the BBB permeability of brain targeting drug candidates has become a critical step in the development of therapeutic agents for CNS diseases.

Conventionally, the BBB permeability of drug molecules has been evaluated with *in vitro* BBB models^{7,8} or directly by animal experiments^{9,10}. Those methods are expensive, time-consuming, labor-intensive and unscalable. In contrast, *in silico* modeling based on machine learning is able to make predictions with good accuracy while largely avoiding the drawbacks of the conventional methods¹¹.

Previously, different types of *in silico* models have been developed for predicting the BBB permeability^{12–17}, including logistic regression¹⁸, support vector machine (SVM)¹⁹, decision tree²⁰, random forest²¹ and light gradient boosting machine (LightGBM)²². Despite the variety of the algorithms involved, the input features for training and prediction have mostly been limited to the structures or the properties of the drug molecules. The most commonly used features are the physicochemical properties, such as the molecular weight, the polar surface area, and the octanol/water partitioning coefficient (LogP)^{23,24}. In particular, Xu et al. introduced a special set of drug features, which combined clinical phenotypes (side effects and indications) of a drug molecule with its physicochemical properties^{25,26}. Additionally, the probability of being a substrate of efflux transporters has also been used as a drug feature in the prediction of BBB penetration^{27–29}.

The previously established models only take the features of each individual drug as the input. However, the BBB penetration of a drug molecule depends not only on the properties of the drug itself, but can also be affected by drug-protein interactions^{11,19}, which play an important role in the mechanisms behind BBB penetration (Figure 1). In the case where a lipophilic drug crosses the BBB through passive diffusion, the BBB permeability is mainly determined by the physicochemical properties of the drug^{23,30}. However, drugs can also be transferred across the BBB by the influx transporters on the surface of the endothelial cells^{31,32}. For example, levodopa³³ and melphalan³⁴ enter the brain through neutral amino acid transporters. In contrast, active efflux transporters such as P-glycoprotein (P-gp) and multidrug resistance-associated protein 1 (MRP1) can pump drugs out of the endothelial cells and thus expel the drugs back to the blood^{2,35}. In addition, protein binding can also influence the BBB permeability of drugs. Drugs bound to plasma proteins such as Albumin tends to stay in the blood because the proteins can hardly cross the BBB^{36,37}; only the unbound fraction of the drugs in the plasma may have a chance to go across the BBB. Besides, structural similarity between drug molecules can also be exploited to infer the BBB permeability of drugs. Structurally similar drug molecules are likely to show similar physicochemical properties and may bind to the same proteins^{38–40}. Thus, the drug-drug similarity is closely related to the factors affecting BBB penetration. Altogether, the drug-protein interactions and the drug-drug similarity are important relations that can contribute to the prediction of BBB penetration. A model that can handle those relations, as well as the features of each drug molecule, is highly desirable.

Graph convolutional networks (GCNs) have been drawing more and more attention during the past few years^{41–48}. In a GCN model, the data is presented in the form of a graph, with the samples as the nodes and the relations between the samples as the edges. The graph convolution operation updates the representation of each node through the aggregation of the messages passed from its

neighboring nodes. Thereby, every node learns about its topological context while preserving its own features. Since graphs can efficiently represent the different types of interactions among drugs and proteins, the GCN model emerges as a promising model for the prediction of BBB permeability. The vanilla GCN treats all relations in a graph as the same. However, different drug-protein relations can have different effects on BBB penetration. For example, drugs related to the influx transporters are expected to be BBB penetrating, whereas drugs binding to plasma proteins are unlikely to cross the BBB. As a result, the vanilla GCN model is not suitable for predicting BBB permeability because it does not distinguish between different types of relations. To solve that problem, a variant of the GCN, which is called relational GCN (RGCN), can be employed since it is able to handle heterogeneous graphs (i.e., graphs containing various types of nodes and edges) by assigning different weight matrices to different kinds of relations^{49–51}.

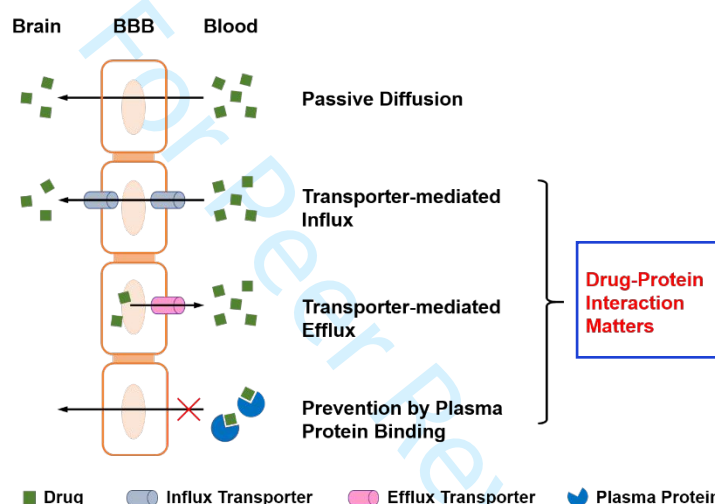


Figure 1. Mechanisms of BBB penetration for small-molecule drugs.

Herein, we developed an RGCN model for predicting the BBB permeability of drug molecules (Figure 2). The model was trained with a graph where the nodes represented drugs and proteins and the edges represented drug-protein/drug-drug relations. Initially, the training was conducted with the drug-protein interactions and the Mordred descriptors⁵² of the drug molecules as the input. The resultant model showed better performance than most of the previously reported ones^{22,53}, highlighting the important role of the drug-protein interactions in predicting BBB permeability. Moreover, the drug-drug similarity was integrated into the RGCN model to further improve the predictive capability of the model. Interestingly, we also found that drug embeddings from the drug repurposing knowledge graph (DRKG)⁵⁴, which encoded the relations between drugs and various pharmacological entities (e.g., genes and diseases), could contribute to the prediction of BBB penetration. Using a combination of the Mordred descriptors and the DRKG embeddings as the drug features led to the best performance of the RGCN model.

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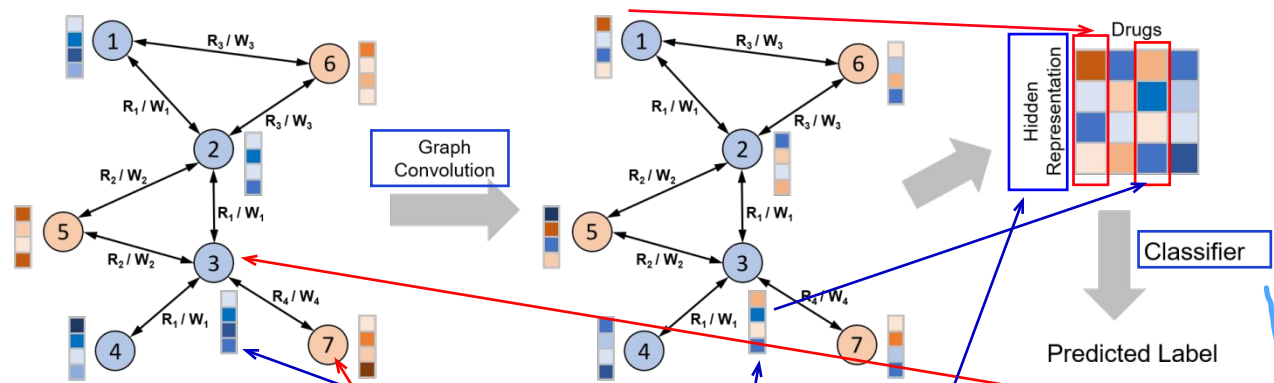


Figure 2. Schematic illustration of the RGCN model. First, the data is structured into a graph. Node 1-4 (blue) represent drugs and Node 5-7 (orange) stand for proteins. All the drug nodes are labeled according to the BBB permeability of the drug. Next to the nodes are the node features. The Mordred descriptors (plus the DRKG embeddings in certain cases) are used as the features of the drug nodes. The features of the protein nodes are treated as learnable parameters. R_1 , R_2 , R_3 , R_4 and R_5 represent the five types of relations between the nodes, i.e., Drug-Protein Influx, Drug-Protein Efflux, Drug-Protein Carrier, Drug-Protein Other and the drug-drug similarity. W_1 , W_2 , W_3 , W_4 and W_5 are the corresponding matrices used for the relation-specific transformation of the feature vectors of the neighboring nodes. The hidden representations of the drugs are obtained through graph convolution and are subsequently passed through two linear layers for classification. Two convolution layers have been used in our experiments.

2. Material and methods

2.1 Data collection

Drugs labeled with their BBB permeability were collected from the literature^{15,21,22,53}. The Simplified Molecular-Input Line-Entry System (SMILES) strings were used to identify the drugs. The drugs with invalid SMILES strings were removed. The duplicates were also deleted. The drug-protein interactions were gathered from STITCH⁵⁵. The interactions with a combined score of 500 or higher were used. We further classified the drug-protein interactions into four categories according to the function of the involved protein. The proteins belonging to the solute carrier (SLC) family usually function as transporters facilitating the uptake of drugs into the brain⁵⁶. The drug-protein interactions involving the SLC family were grouped into one category, which was named Drug-Protein Influx. The ATP-binding cassette (ABC) family contains efflux transporters that inhibit the penetration of drugs across the BBB⁵⁷. And the interactions between the drugs and the ABC transporters were labeled as Drug-Protein Efflux. The carrier proteins bind to certain drugs and modify the pharmacokinetics of the drugs (e.g., BBB penetration)⁵⁸. The interactions involving the carrier proteins constitute the third category, Drug-Protein Carrier. All the other drug-protein interactions fell into the fourth category called Drug-Protein Other. The four types of drug-protein interactions would be handled as four different kinds of relations in the RGCN model.

2.2 Drug features

Mordred descriptors⁵² and DRKG⁵⁴ embeddings were used as the drug features. The 2D Mordred descriptors were generated using the molecular featurizer provided by the DeepChem library⁵⁹. The dimension of the Mordred descriptors was 1613. The DRKG embeddings were obtained by training a TransE₁₂ model with the DRKG, which included 97238 entities from 13 entity-types and 5874261 triplets belonging to 107 edge-types. The DRKG embedding had a dimension of 400.

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The DRKG embeddings were not available for some of the drugs in our dataset because those drugs were just not present in the DRKG. In those cases, learnable parameters were used instead. The drug features were standardized before they were fed to the RGCN model as the node features for the drugs.

2.3 Drug-drug similarity

The drug-drug similarity was evaluated by the Tanimoto score, which was calculated with the 2D pharmacophore fingerprints of the drug molecules³⁸. The pharmacophore fingerprints were generated using the cheminformatics toolkit RDKit⁶⁰. The definitions of chemical features were from the file BaseFeatures provided by RDKit. The chemical feature called ZnBinder was skipped. The distance bins were set to (0, 2), (2, 4), (4, 6), and (6, 10). Two-point and three-point pharmacophores were identified. And the fingerprints were generated accordingly. The threshold of the similarity score was optimized as a hyperparameter.

2.4 RGCN model

The RGCN is an extension of the classic GCN and shares the same essential algorithm with the GCN^{49,50}. In the GCN model, all the feature vectors of the neighboring nodes are transformed with the same weight matrix. In contrast, for the RGCN, the concept of relation types has been introduced. Herein, the vector transformation is relation-specific, i.e., each type of relation is assigned a distinct weight matrix. As a result, the RGCN model is able to handle heterogeneous graphs.

In the present work, an RGCN model was built using Pytorch Geometric⁶¹. The RGCNConv module provided by Pytorch Geometric was directly used as the convolution layer in our model. There were two convolution layers followed by two linear layers. When the Mordred descriptors were used as the drug features, the sizes of the input and the output feature vectors were 1613 and 1600 for the first RGCN layer, 1600 and 900 for the second RGCN layer, 900 and 400 for the first linear layer, and 400 and 2 for the second linear layer. When the Mordred descriptors were combined with the DRKG embeddings, the initial input feature vector had a size of 2013. The sizes of the output feature vectors for the four layers were 2000, 1000, 500 and 2, respectively.

2.5 Model training and evaluation

The data collected were structured into a graph. The drugs were split into the training set and the testing set at a ratio of 4:1 in a stratified manner. The Adam optimizer and the negative log-likelihood loss (NLLoss) were used for the training. 10-fold cross-validation was first conducted with the training set for hyperparameter tuning. After optimization, the learning rate was set to 0.0005, and the weight decay of the Adam optimizer was set to 0.01. The dropout rate was set to 0.5 for all experiments. The model was then trained with the entire training set and evaluated with the testing set.

The performance of the model was assessed by multiple metrics, including the accuracy, the sensitivity (the true positive rate), the specificity (the true negative rate), the area under the receiver operating characteristic (AUROC), and the area under the precision-recall curve (AUPRC). The scores were calculated with the validation set during the cross-validation or with the testing set during the final testing step. The AUPRC was obtained by computing the average precision. Particularly, the average precision was calculated with the nonpenetrating drugs as the positive

class. The baseline value of the AUPRC was 0.297, which was obtained by dividing the number of the nonpenetrating drugs (1178) by the total number of the drugs in the dataset (3961), given that the data was split in a stratified manner.

3. Results

3.1 RGCN model featuring the drug-protein interactions

An RGCN model was first developed to demonstrate the role of drug-protein interactions in the prediction of BBB permeability.

A graph was prepared with 3961 drugs and 4264 relevant proteins as the nodes. All the drug nodes were labeled either as penetrating drugs (2783) or nonpenetrating drugs (1178). Edges were established according to the drug-protein interactions. There were 25021 drug-protein edges, among which 1007 were for Drug-Protein Influx, 308 were for Drug-Protein Efflux, 417 were for Drug-Protein Carrier, and 23289 were for Drug-Protein Other. The Mordred descriptors, which represent the structural and the physicochemical properties of the drug molecules, were introduced as the node features for the drugs. The node features for the proteins were treated as learnable parameters.

The RGCN model was trained and evaluated with the prepared graph. The results from 10-fold cross-validation with the training dataset using the optimized hyperparameters are shown in Table 1. The model gave an overall accuracy of 88.1%, with a sensitivity of 0.940 and a specificity of 0.741. The AUROC was 0.897. Since the dataset was imbalanced, the AUPRC was also calculated (with the nonpenetrating drugs as the positive class) to evaluate the model. The result was 0.834, which was significantly higher than the baseline value 0.297 (Section 2.5). The performance of the RGCN is among the best results reported in recent papers. In particular, we employed the LightGBM as the base model for comparison because it is one of the most powerful models for tabular data. Na et al. lately reported a LightGBM model that showed top performance in predicting BBB permeability²². However, the high performance was achieved partially due to data leakage, which was caused by the duplicates in their dataset. Canonical SMILES strings instead of the raw SMILES strings should have been used to identify the drugs. Therefore, we trained a LightGBM model ourselves. And the model was rigorously optimized using Optuna⁶², which is an automatic hyperparameter optimization framework. As shown in Table 1, the RGCN model showed higher overall accuracy compared to the LightGBM model, although with the lower AUROC and AUPRC scores. It is worth noting that significantly higher specificity was obtained with the RGCN model, which indicates that the RGCN model is much better at correctly classifying the nonpenetrating drugs. The higher specificity results from the capability of the RGCN to learn the topological structure of the graph, i.e., the drug-protein interactions.

Table 1. Performance of the RGCN model with only the drug-protein interactions as the edges.

Model	Experiment	Accuracy	Sensitivity	Specificity	AUROC	AUPRC
RGCN	Cross-Validation	0.881	0.940	0.741	0.897	0.834
	Testing	0.875	0.921	0.767	0.904	0.810
LightGBM	Cross-Validation	0.868	0.951	0.670	0.910	0.845
	Testing	0.870	0.943	0.699	0.925	0.837

3.2 Drug-drug similarity boosting the performance of the RGCN model

Drugs with similar structures tend to show similar properties and thus have similar BBB permeability. Introduction of drug-drug similarity to the RGCN can enhance the networking between the drug nodes and facilitate the message passing within the graph, leading to better performance of the model.

Herein, the drug-drug similarity was added to the graph as the fifth type of node relation. Edges were established between two drugs if the similarity score (Section 2.3) was higher than the similarity threshold. The similarity threshold determined the number of the drug-drug edges. And it was optimized as a hyperparameter. Table 2 shows the performance of the RGCN model with the similarity threshold set to different values. The model got the highest overall performance when the similarity threshold was set to 0.7, indicating that a balance had been reached between the number of edges and the level of similarity. In this case, 15167 drug-drug edges were integrated into the data graph. The similarity threshold was kept at 0.7 for all the further experiments.

Table 2. Optimization of the similarity threshold.

Similarity Threshold	Number of Edges (Drug-Drug Similarity)	Accuracy	Sensitivity	Specificity	AUROC	AUPRC
0.9	5620	0.883	0.938	0.752	0.913	0.847
0.8	8834	0.874	0.911	0.786	0.917	0.828
0.7	15167	0.888	0.944	0.756	0.913	0.856
0.6	25770	0.879	0.918	0.786	0.917	0.819
0.5	48119	0.877	0.911	0.795	0.910	0.808

The performance of the RGCN model was boosted after drug-drug similarity was introduced to the graph (Table 1 and Table 3). All the cross-validation scores were now higher than those for the LightGBM model except for the sensitivity. However, with regard to the results from the testing experiments, the AUROC and the AUPRC of the RGCN model were still lower compared with the base model.

Table 3. Performance of the RGCN model with drug-drug similarity added. The similarity threshold was 0.7.

Model	Experiment	Accuracy	Sensitivity	Specificity	AUROC	AUPRC
RGCN	Cross-Validation	0.888	0.944	0.756	0.913	0.856
	Testing	0.881	0.914	0.805	0.917	0.818
LightGBM	Cross-Validation	0.868	0.951	0.670	0.910	0.845
	Testing	0.870	0.943	0.699	0.925	0.837

3.3 DRKG embeddings contributing to the prediction of BBB permeability

In addition to drug-protein interactions and drug-drug similarity, the relations between drugs and other pharmacological entities (e.g., phenotypes) may also contribute to the prediction of BBB penetration²⁵.

DRKG is a comprehensive biological knowledge graph that relates genes, compounds, diseases, biological processes, side effects and symptoms. The DRKG embeddings encode the relations

between all of those pharmacological entities. Herein, the DRKG embeddings were combined with the Mordred descriptors to bring extra information about pharmacological interactions.

As shown in Table 4, adding the DRKG embeddings to the RGCN didn't raise the evaluation scores in the cross-validation results. But it did notably improve the AUROC and the AUPRC scores from the testing experiments (Table 3 and Table 4). Now all the testing scores of the RGCN model except the sensitivity were higher than those of the LightGBM model.

Table 4. Performance of the RGCN model when combined drug features consisting of the Mordred descriptors and the DRKG embeddings were used. Both the drug-protein interactions and the drug-drug similarity were included in the data graph.

Model	Experiment	Accuracy	Sensitivity	Specificity	AUROC	AUPRC
RGCN	Cross-Validation	0.885	0.943	0.746	0.905	0.850
	Testing	0.873	0.907	0.792	0.928	0.840
LightGBM	Cross-Validation	0.868	0.951	0.670	0.910	0.845
	Testing	0.870	0.943	0.699	0.925	0.837

4. Discussion

The RGCN gains advantage over the previous models by handling the drug-protein interactions. Thus, it is more effective in making predictions for the drugs whose BBB permeability depends on the drug-protein interactions. However, those drugs are just a small proportion of all the drugs in the BBB dataset. Currently, most of the drugs being studied in the field of BBB penetration are supposed to cross the BBB through the mechanism of passive diffusion. As a result, the RGCN can only improve the prediction for the whole dataset to a limited extent.

The datasets used for predicting BBB penetration are usually imbalanced. Therefore, the AUPRC is a more appropriate metric for the model evaluation. In the present work, the model was trained by minimizing the NLLoss. And the AUPRC was calculated with the predicted probabilities and the true labels. Recently, Yang et al. reported a principled technical method to optimize AUPRC for deep learning⁶³. With this method, the authors were able to train their models by directly maximizing the AUPRC and achieve better results than using other approaches. We consider the direct AUPRC optimization to be a promising strategy to enhance the prediction of BBB penetration and will employ it in our future work.

5. Conclusion

We have successfully established an RGCN model for the prediction of BBB permeability. In our study, different mechanisms of BBB penetration for small-molecule drugs were considered. For each mechanism, the factors that affect the BBB permeability were identified. And the RGCN model was chosen to handle all the factors involved, i.e., the drug properties, the drug-protein interactions, and the drug-drug similarity. Therefore, our model can make predictions for small-molecule drugs penetrating the BBB through all kinds of mechanisms, while the previously built models only apply to lipophilic drugs that penetrate the BBB through passive diffusion. The RGCN model represents a new strategy for predicting BBB permeability and can potentially serve as a generic platform for the prediction of other pharmacological properties of drug molecules.

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For Peer Review

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